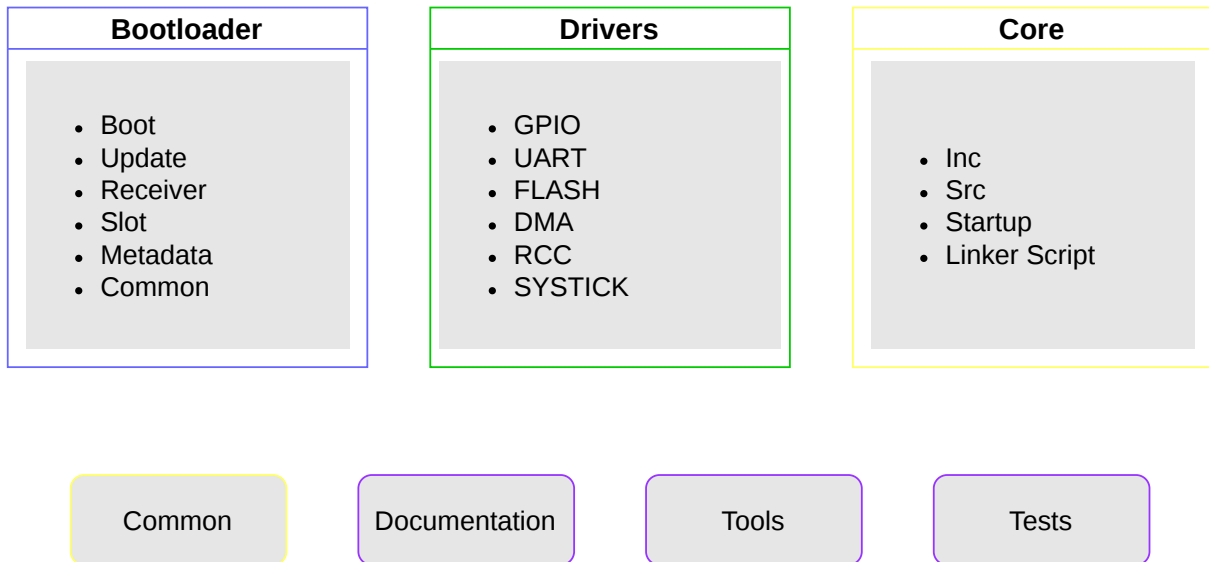


Repository Structure

How is the repository organized?

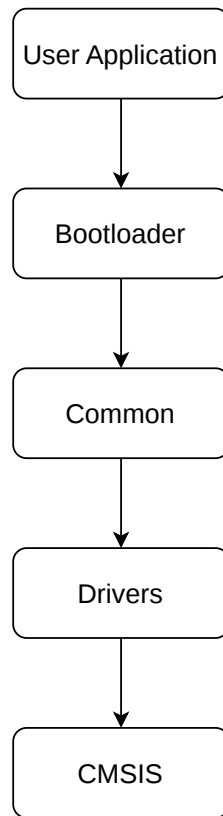
HEXBOOT_F0



NOTE: The repository is organized into independent modules to improve maintainability and scalability.

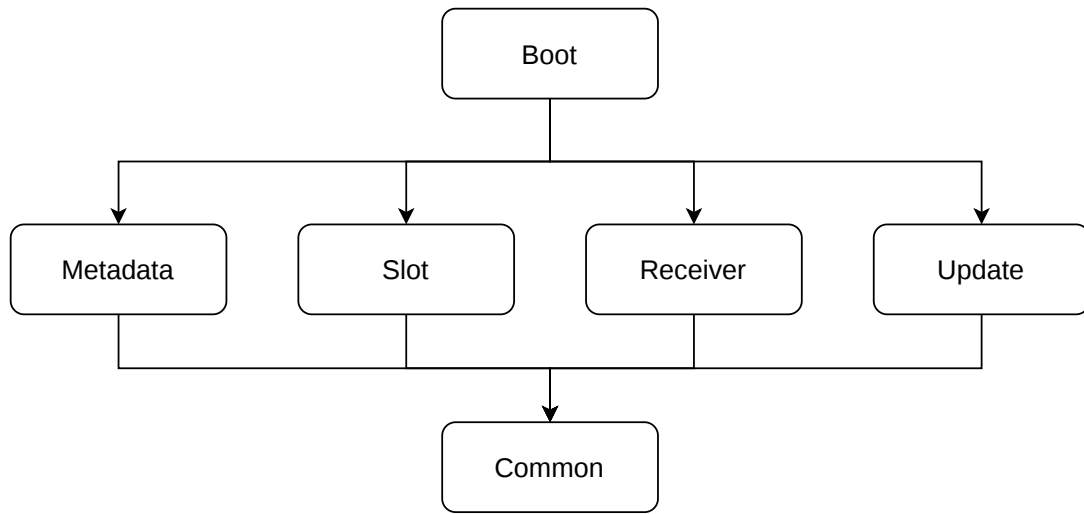
Layer Dependency

Who depends on whom?



Module Dependency

How are the bootloader modules related?



NOTE:

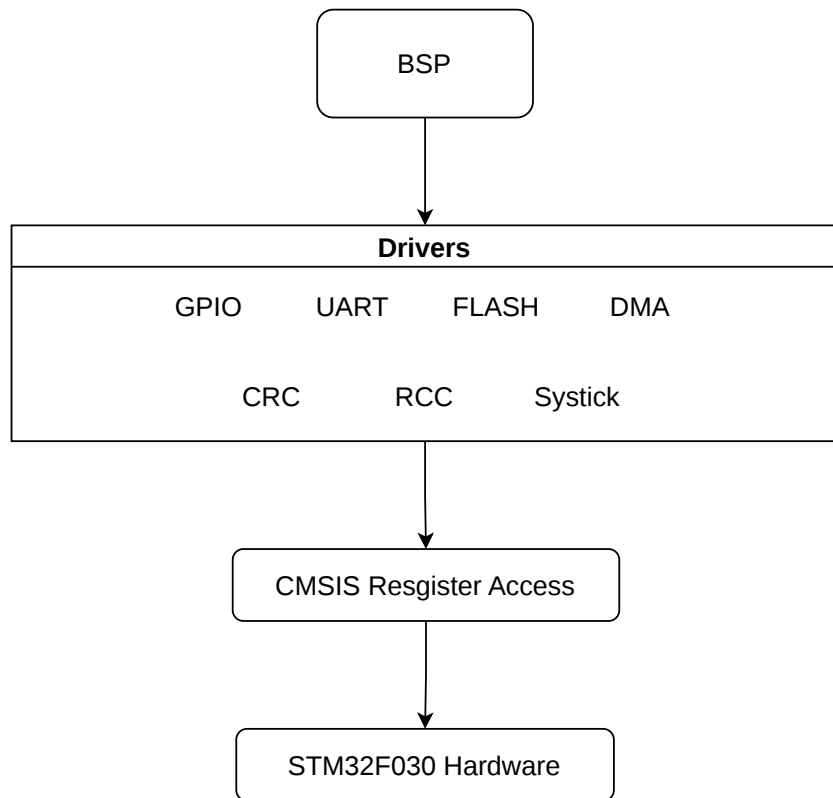
Shared functionality is provided through the Common module.

Module Responsibilities

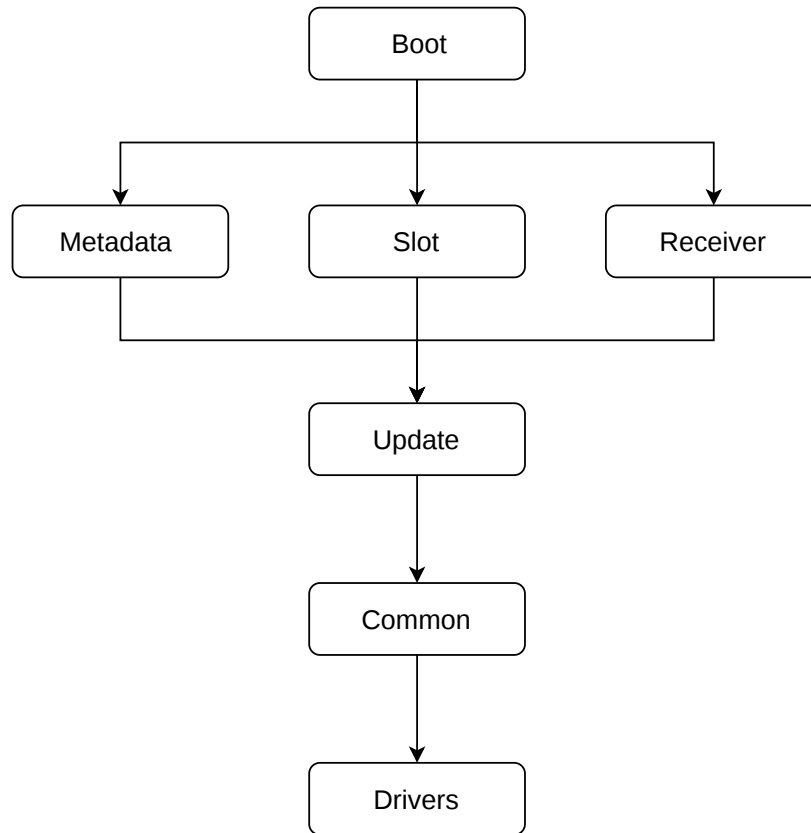
which module is responsible to perform which task

Module	Responsibility	Depends on
Boot	• Boot decision • Startup sequence • Jump to application	• Metadata • Slot
Metadata	• Read metadata • Write metadata • Validate metadata	• Common
Slot	• Select active slot • Select update slot • Validate slot	• Metadata • Common
Receiver	• Receive firmware • Parse packets • Buffer data	• Common
Update	• Program Flash • Verify firmware • Coordinate update	• Receiver • Metadata • Common
Common	• Shared macros • Utility functions • Common data types	

Driver Architecture

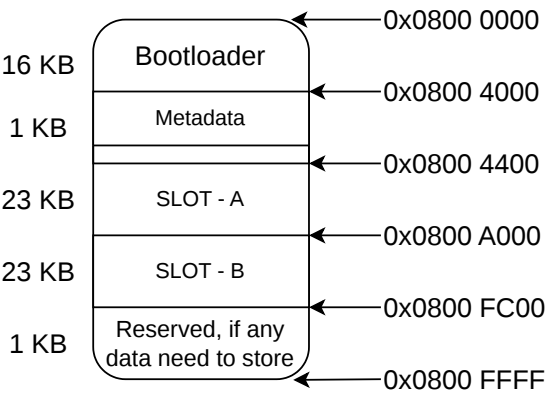


Bootloader Architecture



Memory Layout

STM32F030C8T6



Coding Rules

Rule	Description
Naming	snake_case for functions
Globals	Avoid global variables
Driver API	Public API in .h, implementation in .c
Register Access	CMSIS only
Memory	No dynamic allocation
Error Handling	Return status codes